

Emmanuel Niyi-Oriolowo, MBBS, BSc

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PROFESSIONAL SUMMARY

Medical doctor with strong software engineering experience and an applied machine learning focus in healthcare. Experienced in epidemiological data engineering, time series forecasting, and diagnostic classification using clinical datasets. Seeking advanced training in health data science and analytics to develop responsible, data-driven solutions.

EDUCATION

Newcastle University 2026 – present

MSc Advanced Data Science with Health

Bowen University 2016 – 2024

MBBS (2024), BSc Physiology (Intercalated, 2020)

TECHNICAL SKILLS

- **Programming & Development:** Python (Django, FastAPI), SQL (PostgreSQL, MySQL), RESTful APIs, Docker, Pytest, Git/GitHub
- **Machine Learning & AI:** Scikit-learn, FastAI, Pandas, NumPy, feature engineering, model evaluation, Jupyter
- **Data Analysis & Visualization:** Matplotlib, statistical analysis, exploratory data analysis, data cleaning and preprocessing
- **Cloud & DevOps Tools:** AWS and Azure (basic), Linux environments, Postman
- **Healthcare Domain Knowledge:** Clinical diagnostics and patient management, medical terminology, medical coding systems, EHR workflows, patient data privacy principles

MACHINE LEARNING AND DATA SCIENCE PROJECTS

NCDC Lassa Fever Weekly Time Series Dataset (Nigeria, 2020–2025)

Epidemiology and Data Engineering December 2025

- Created a cleaned and standardized weekly time series dataset of Lassa fever cases from NCDC epidemiological reports published as PDFs.
- Built a reproducible extraction and data-cleaning pipeline converting inconsistent reports into machine-readable formats.
- Published the dataset on Kaggle and Hugging Face with documentation of data gaps and limitations.
- Dataset adopted for ongoing doctoral research with supervising faculty approval.

Data-Driven Surveillance: Lassa Fever Forecasting for Public Health Action

Healthcare Predictive Modeling December 2025

- Built and evaluated ETS, SARIMA, and machine learning models for weekly Lassa fever forecasting.
- XGBoost achieved the strongest short-term performance among evaluated approaches.
- Used feature importance and SHAP analysis to identify seasonal effects and recent case momentum.

Replication of Liver Disease Classification with Data Augmentation

Healthcare Machine Learning October 2025

- Replicated a published study using SMOTE and GAN-based augmentation.
- Introduced Matthews Correlation Coefficient to assess robustness under class imbalance.
- Demonstrated that GAN-based augmentation inflated performance, while SMOTE improved reliability.

CT Hematoma Classifier – Multi-Class

Medical Imaging and Deep Learning September 2025

- Built a multi-class deep learning model for intracranial hematoma classification on CT scans.

- Evaluated performance on internal and external datasets.
- Applied Grad-CAM to assess alignment with clinically relevant image regions.

PUBLICATIONS

Polygenic Risk Scores for Cardiovascular Disease: Clinical Utility and Limitations. Adejumo, F.A., Obielodan, O.O., Egwu, E.O., et al. *The Egyptian Heart Journal*, 2026. DOI: 10.1186/s43044-026-00746-3

PROFESSIONAL EXPERIENCE

Medical Doctor *October 2024 – October 2025*
Federal Medical Centre Abuja

- Delivered supervised clinical care in pediatrics, surgery, internal medicine, and obstetrics and gynecology.
- Managed high-acuity cases in Accident and Emergency settings, contributing to acute decision-making and multidisciplinary resuscitation.
- Assisted in surgical procedures and supported pre-operative optimization and post-operative recovery.
- Supported infection prevention audits and quality improvement initiatives focused on patient safety.
- Improved patient documentation workflows and facilitated bedside teaching during ward rounds.

Backend Developer *September 2024 – March 2025*
KDI Consultancy Ltd

- Designed and implemented backend architecture for a currency exchange MVP supporting secure authentication and concurrent transactions.
- Built asynchronous processing pipelines using Celery and Redis to enable real-time transaction handling.
- Achieved high test coverage through unit and integration testing across core services.
- Developed an intelligent chatbot using LangChain to answer user queries and explain service workflows.
- Collaborated with frontend engineers to ensure consistent API design and smooth system integration.

Backend Developer Intern *December 2023 – September 2024*
KDI Consultancy Ltd

- Built document ingestion and validation pipelines across multiple formats to automate compliance checks.
- Automated workflows aligned with PCI-DSS and ISO 27001 standards, reducing manual review time.
- Implemented secure authentication and role-based access control for sensitive compliance data.
- Achieved full test coverage across core modules through structured testing.

Backend Developer Volunteer *March 2021 – September 2023*
Ontriv LLC

- Led development of a multi-platform social media scheduling tool automating content publishing.
- Integrated platform APIs and implemented secure workflows for authentication and media uploads.
- Engineered the scheduling engine with retry logic and error handling for reliable delivery.

CERTIFICATIONS AND CONTINUOUS LEARNING

- Future of AI, BlueDot Impact (2026)
- Human Research (Data or Specimens Only Research), Citi Program (2026)
- Kaggle Courses in Python, Machine Learning, Pandas, and Data Visualization (2025)
- American Heart Association Basic Life Support Certification (2019)

LEADERSHIP AND ACHIEVEMENTS

- Deputy House Officers' Representative, Federal Medical Centre Abuja (2024–2025)
- Coordinator, 2025 Blood Drive, Federal Medical Centre Abuja (80+ units collected)
- Class Representative, Bowen University Medical Class of 2024 (2016–2024)